

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 17

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm²) • They use the polyp's CO₂ and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently • They supply the polyp with glucose and oxygen - over 95% of its food energy; the polyp provides a protected habitat and nutrients • Polyps also trap plankton with cilia and stinging nematocysts; surplus CO₂ is used to build the calcium carbonate skeleton	4
01.2	• High light (shallow, clear water) for the algae; warm stable temperatures (about 18-30 °C); low turbidity; constant salinity; well-oxygenated water • Found only where all these coincide (about 1% of ocean area) • Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance	3
01.3	<i>Any 3 from:</i> • Fisheries: support over a quarter of marine species including fish, lobsters, conch • Erosion protection: complex structure dissipates wave energy protecting low coral islands; coral sand builds shores • Medicinal discoveries: AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts • Climate control: carbonate skeletons sequester CO₂ • Tourism (diving, snorkelling)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
02.1	• EDGE: Evolutionarily Distinct and Globally Endangered - few close relatives, often the last species of their genus / family (Bactrian camel, pygmy hippo, northern bald ibis); loss would remove a unique branch of evolution / genes • Endemic: found in only one area (often islands - Galapagos, Madagascar, Seychelles: Aldabra giant tortoise, red ruffed lemur); if lost locally there is no population elsewhere to recolonise • Keystone: disproportionately important for the structure of the community relative to its abundance - predation of dominant species, seed dispersal, habitat creation (jaguar, forest elephants keeping paths and water holes open, wolves, beavers); losing them causes loss of many other species • Flagship: high public profile (tiger, panda, elephant, orang-utan) raising funds and support that also protects the whole habitat and less charismatic species (fungi, bats, moths)	8
02.2	• A fragmented population may consist of isolated groups each too small to be viable (resources, inbreeding) even if total numbers are large • A species confined to a single area (eg bonobos in one country) is vulnerable to any local change / conflict, unlike chimpanzees spread across many countries	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 28-29; AQA spec 3.1.2.3.1.	
03.1	• Plants close their stomata to reduce transpiration losses • Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop • Less water for nutrient absorption / transport and cell turgidity; severe shortage dehydrates cells and inhibits biochemical reactions	3
03.2	• Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms; higher risk of fungal disease; root respiration inhibited • Drainage ditches or pipes • Deep ploughing; avoiding compaction by machinery / livestock; adding organic matter to encourage worms	4
03.3	• Groundwater often contains dissolved salts; when the water evaporates from the soil the salts are left behind and accumulate • High soil salinity causes osmotic dehydration of roots / crop death; washing out with extra water uses more water and salinises rivers	2
03.4	• Add crushed lime (calcium carbonate / hydroxide); low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	
04.1	• k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses • Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population • Deep-water aggregations easily located and caught by trawling • Recovery takes decades so fisheries have had to be closed	4
04.2	• Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young • Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed	2

Question	Marking guidance	Marks
04.3	• Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort • Masks the decline of the breeding population until a sudden collapse	2
04.4	• Neither is a target species but both are caught in demersal trawls; skate now extinct in the Baltic and critically endangered elsewhere • Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	
05.1	• Negative • Positive • Negative • Positive	4
05.2	• The response to a change acts to re-establish / return the system to the previous equilibrium • So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback	2
05.3	• eg the possible release of methane hydrate at the end of the Permian era • Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium	2
05.4	• Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) • Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
06.1	• Hard release: individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) • Soft release: gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds • Problems faced: finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups	3
06.2	<i>Any 5 from:</i> • Large enough suitable habitat for a viable population • Reliable food supply • Low predation risk / original threat removed • Suitable breeding sites • Water • Support of the local human population • Official / legal protection of habitat and species	5
06.3	• Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed • Identifies causes of failure so future releases can be improved / extra support given	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	
07.1	• Cycling of materials (like biogeochemical cycles) - recycling, re-use • Energy from renewable sources • Human activities should support ecosystems / conserve biodiversity • Separation of technical and biological materials; connected systems where the waste of one process feeds another (industrial symbiosis) • Diverse systems are more resistant to change • Design of products for extended use and end-of-life re-use; optimum rather than maximum production; technologies that improve efficiency and enable dismantling	6
07.2	• Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae • Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs	2
07.3	• Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them • A common approach gives a level playing field, coordinated monitoring and support for less successful countries	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	
08.1	• $\Sigma n(n - 1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ • $N(N - 1) = 100 \times 99 = 9900$ • $D = 9900 / 2900 = 3.41$ (accept 3.4)	3
08.2	• Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B • The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
09.1	• Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity • Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) • Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop	3
09.2	• A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) • They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) • Many are in regions where environmental degradation / habitat loss threatens the wild plants	3

Question	Marking guidance	Marks
09.3	• A large population may all descend from a few closely related individuals so they are genetically similar • Similar adaptations mean all are susceptible to the same change (disease, climate) • Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs); so species should be protected across their whole range to retain regional genes	3
09.4	• Morama bean (drought resistant, soya-like protein); yeheb tree (drought resistant nuts, poor soils); potato bean (high protein); perennial maize / Kernza (no annual replanting, less ploughing / erosion); salt-tolerant spinach relatives	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes • Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time • Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management • Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
10.2	• Any taxonomic group of organisms classified together by similarity (eg species, genus, family)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	
11.1	Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Marine productivity: light, nutrients, upwelling; where fisheries occur • Methods: demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • Impacts: overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • Population dynamics and MSY: Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • Management: quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear; EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover); role of aquaculture; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	25

Question	Marking guidance	Marks
11.2	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides - Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer - Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms - Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases - Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents - Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25